

Real-time Holographic Media System

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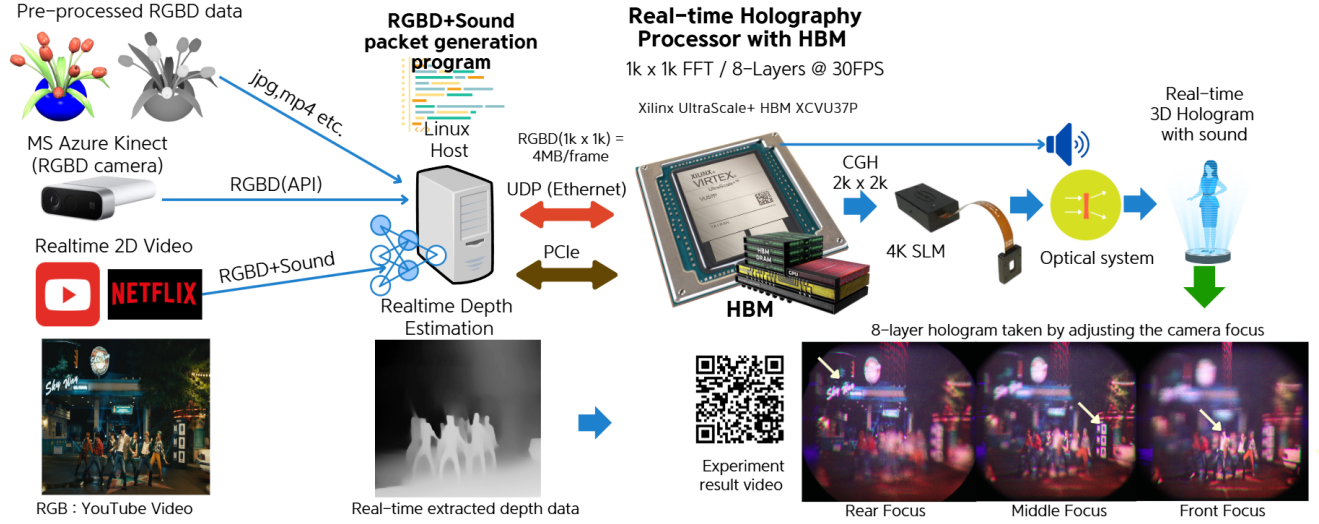


Figure 1: The real-time holographic media system workflow. RGB input videos, such as YouTube content, are transmitted as RGBD+Sound (RGBDS) packets to the Real-time Holography Processor (RHP) through a real-time packet generation program. The RHP generates 4K, 30fps, 8-layered off-axis holograms in real time and sends them to the optical system. A demo video can be accessed via the provided QR code link.

Abstract

This paper introduces a real-time holographic media system that converts 2D or RGBD videos into 3D holograms. The core of this system includes a Linux host that extracts depth information from 2D images and transmits it via packets, and a holography processor leveraging high-bandwidth memory (HBM) to generate volumetric holograms with 8-layer depths at 30 frames per second (fps). Demonstrations highlight the potential of the proposed system for advancing future holographic media technologies.

Keywords

hologram, CGH, holography processor, real-time holographic media

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1 Real-time holographic media system features

This technology enables the real-time conversion of 2D media displayed on a computer screen into 3D holograms. Computer-Generated Holography (CGH) employs layered algorithms to represent depth in three dimensions [Park 2017]. While increasing depth layers enriches the conveyed information, it also significantly raises computational demands. Recent advances in GPU performance allow for real-time CGH generation via parallel programming. However, these solutions are often impractical due to their high costs and power consumption. In contrast, dedicated hardware, such as FPGA (Field-Programmable Gate Array), efficiently calculates CGH with reduced power consumption and a compact form factor. The best existing real-time CGH processors generate 8-layer holograms at 30fps using 4K spatial light modulators (SLMs). However, the FFT processing size and speed are the main bottlenecks, and it is limited to inputting pre-prepared RGBD data, making it difficult to use as a general-purpose real-time holography processor [Kim et al. 2019]. The proposed system overcomes these limitations by employing high-speed, multi-channel FFT using High Bandwidth Memory

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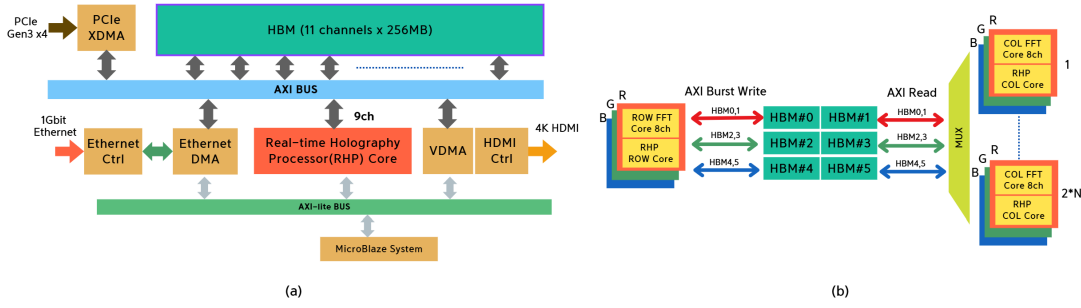


Figure 2: (a) RHP system configuration diagram: A single-chip SoC solution integrating IPs for RHP core, HBM, PCIe, Ethernet, and HDMI using an AXI interface. (b) HBM memory architecture of the RHP, designed to address column FFT read latency.

(HBM). It enables the real-time conversion of general 2D videos, such as YouTube or Netflix content, into 3D holograms. Figure 1 illustrates the system configuration, including input/output images. The system leverages high-speed HBM with multi-channel double buffering to minimize column-wise FFT read latency. Additionally, an off-axis algorithm is integrated into the RHP architecture to eliminate DC and conjugate noise in the reconstructed hologram [Kwon et al. 2024]. Figure 2 outlines the hardware architectures of the RHP and HBM memory. The RHP core block incorporates user-defined hardware for CGH generation and features an AXI interface to manage nine HBM channels. Of these, six channels are dedicated to 2D FFT operations, with pre/post-processing supported by HBM 0 to 5. These HBM channels employ a double-buffering architecture, where two channels per RGB layer compensate for processing speed discrepancies between row and column cores. While row FFT operates efficiently due to contiguous data writes, column FFT suffers from slower performance caused by discontinuous address reads. To mitigate this, the column processing logic has been doubled ($N=1$). The row and column cores employ 24 FFT cores ($\text{RGB} \times 8$ layers) and 48 FFT cores ($\text{RGB} \times 8$ layers $\times$ dual processing), respectively. Key features of real-time holographic media system:

- FPGA-based Real-time Holography Processor (RHP) leveraging multi-channel HBM memory and efficient algorithms.
- Fixed-model CGH generation architecture utilizing Fresnel propagation equations and layered algorithms.
- Full-color volumetric holograms with 8-layer depth at 30fps using 4K SLM.
- Low processing latency (28ms from RGBD to CGH).
- Off-axis hologram support for DC noise removal.
- RGBD + Sound packet transmission software and real-time depth estimation from 2D videos.

2 Summary

In this product, we present a real-time holographic media system capable of generating 4K resolution, 8-layer depth holograms at 30 fps using 2D video input without depth information. For real-time multilayer CGH generation, we use high-speed HBM double buffering technique in the memory of the hologram processor to reduce

the latency during hologram calculation. Additionally, the proposed system can convert any 2D video into holograms in real time with low power consumption and a compact form factor, making it suitable for various applications such as holographic telepresence, holographic video streaming, and automotive holographic head-up displays.

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